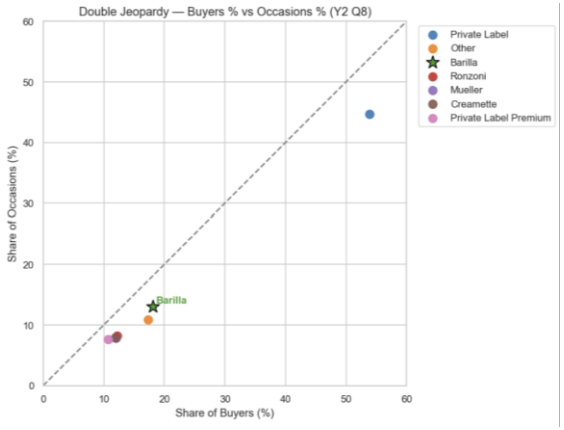
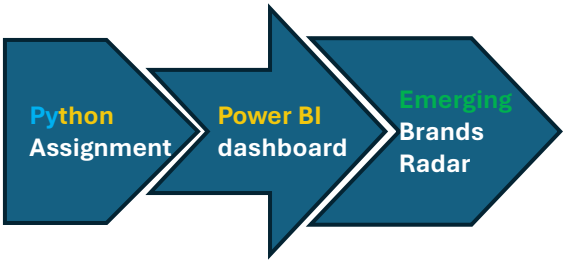
Data projects don't end → they evolve.



Growth comes from **penetration** + **availability**, not loyalty schemes.



Dunnhumby Carbo-Loading (Pasta subset) | Python → Power BI

Python • SQL • Power BI • Marketing Science (NBD–Dirichlet / Double Jeopardy)

One project. Three versions. Still evolving.

Dunnhumby Carbo-Load | 5.2M transactions | Pasta subset

01 Assignment (Python)

Barilla deep dive using marketing science benchmarks.

02 Replica (Power BI)

Rebuilt as an interactive dashboard (sliceable KPIs)

03 Extension (Power BI)

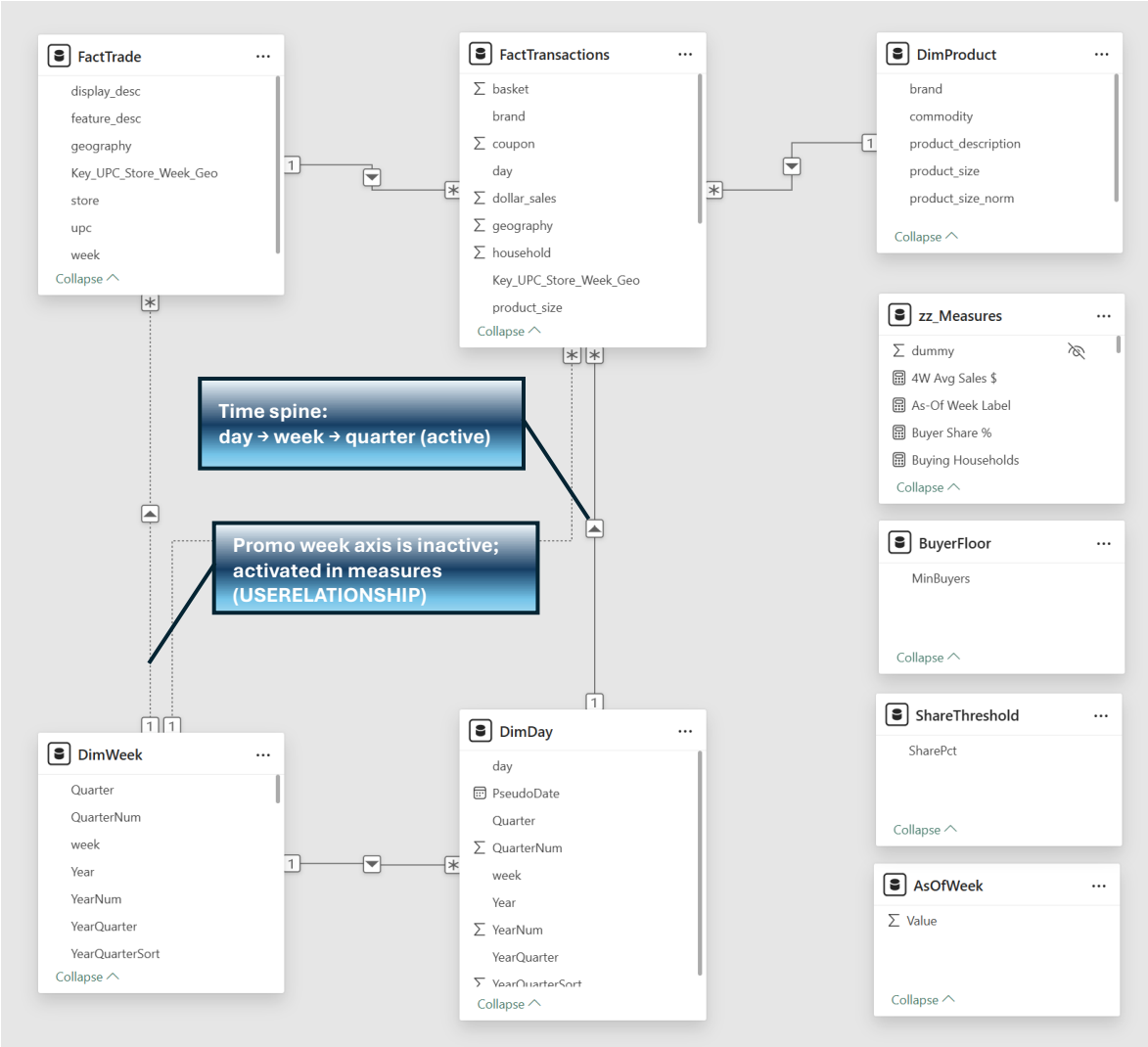
Emerging Brands Radar: persistence + noise controls

- Category stable
- Growth = penetration (reach more buyers)
- Promos = short-term lift
- Loyalty scales with size (Double Jeopardy)

Relational model + time spine + promo linking

(filter-safe by design)

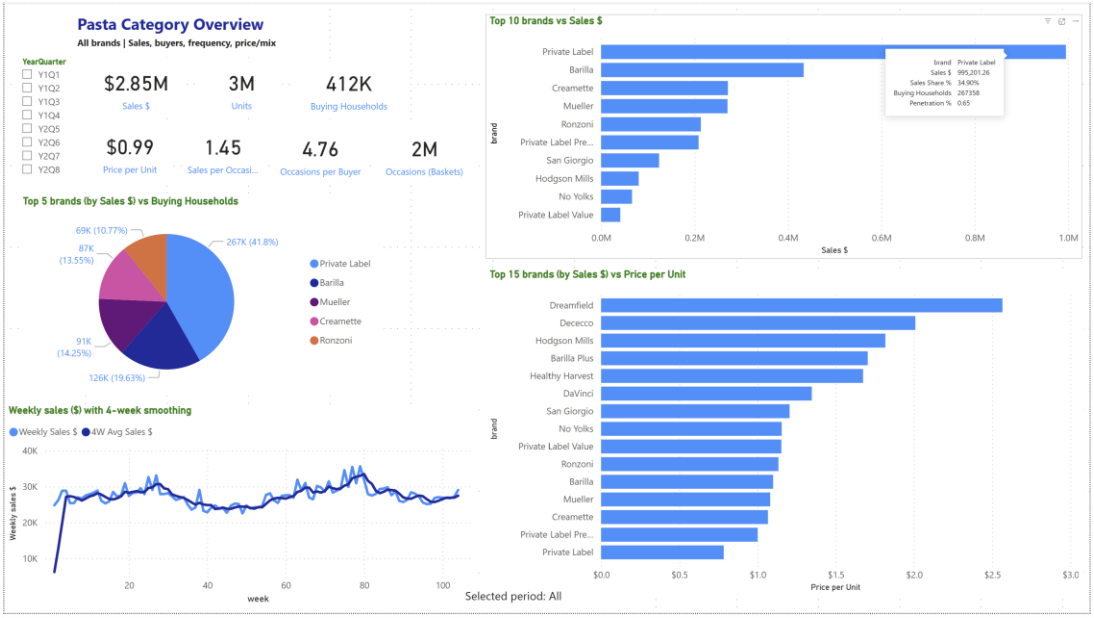
- **Core KPI spine:** FactTransactions ← DimProduct, DimDay
- **Promo linking:** FactTrade joins via composite key
- **Promo time control:** DimWeek→FactTrade inactive; used in promo measures
- **Parameter tables (disconnected):** thresholds & As-Of selection



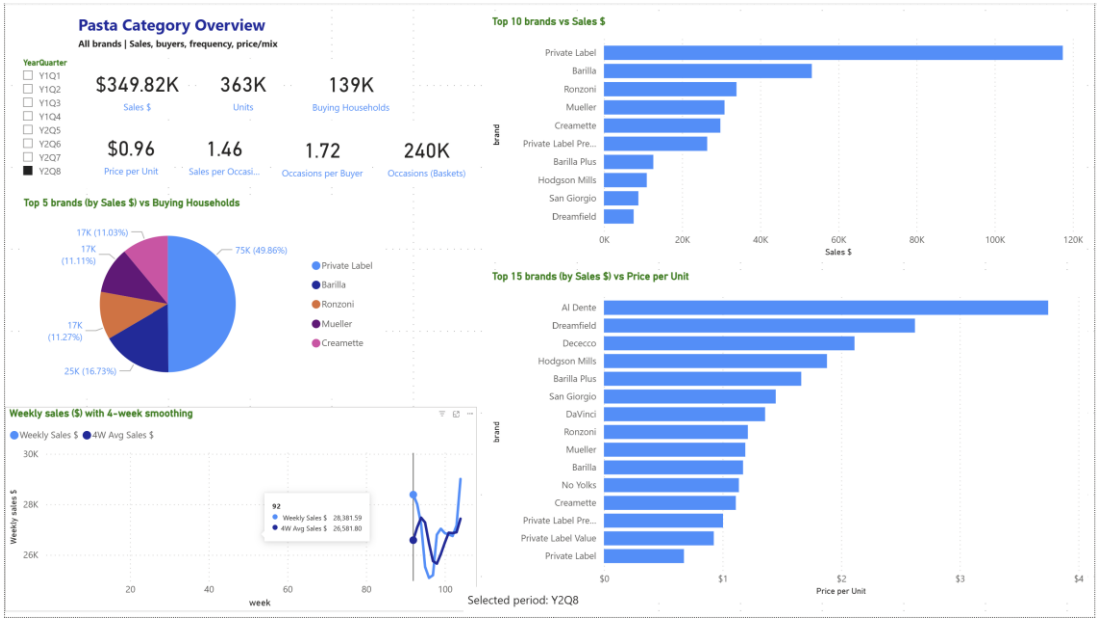
Category Overview: All periods + Y2Q8

Stable category; competition is about penetration, not ‘loyalty magic’.

All periods

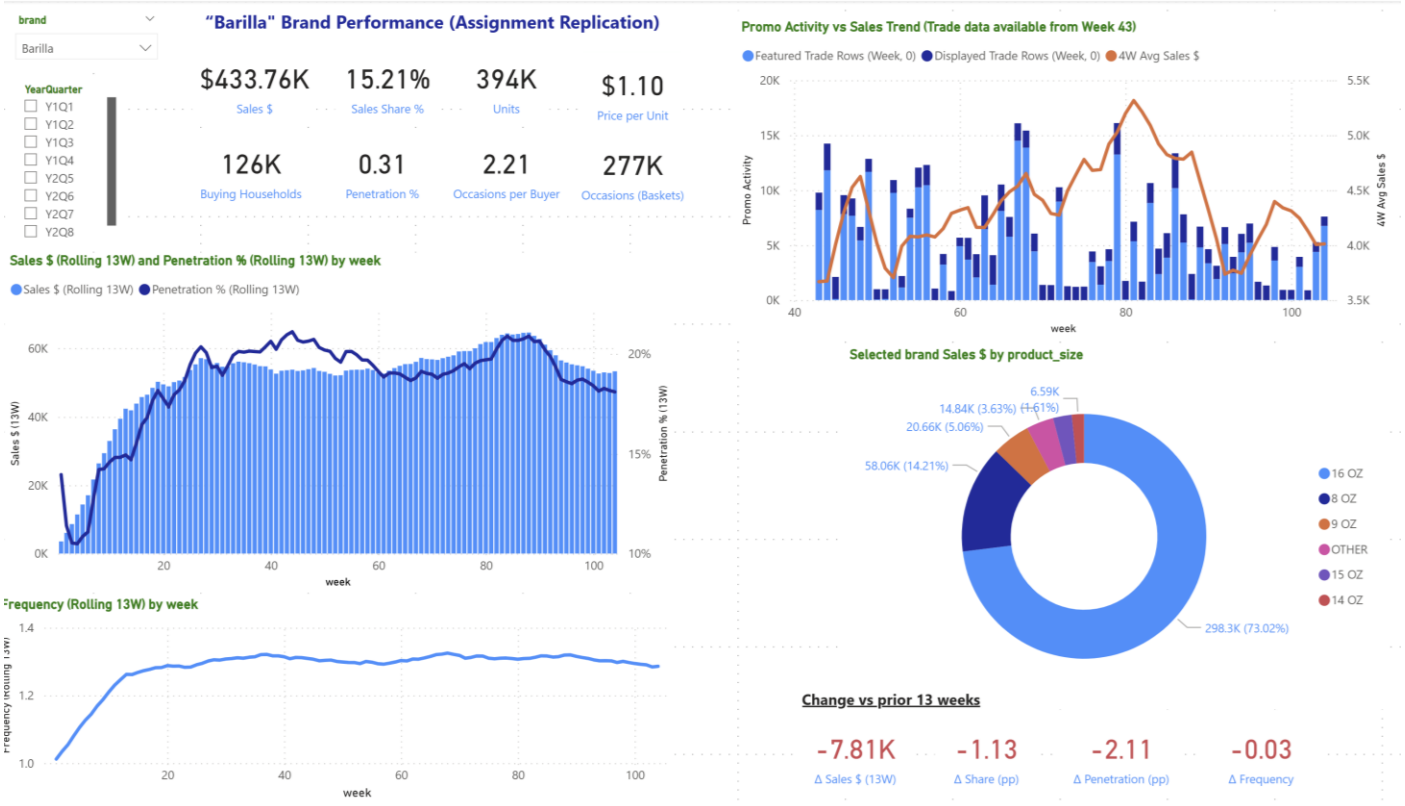


Y2Q8 (latest quarter)



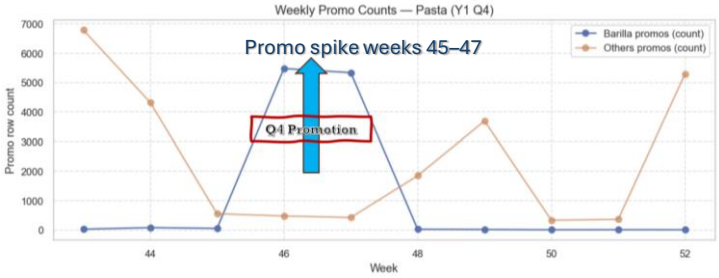
Barilla replica (Power BI)

Same data. Same conclusions. Now it's a dashboard you can use weekly.



- **KPIs confirmed:** 15.21% share, 126K buyers, 2.21 occasions/buyer
- **Double Jeopardy visible:** frequency ~1.2–1.3 across most weeks
- **Promo spike at a glance:** Y1Q4 feature surge → short lift → retreat
- **New insight:** 16 OZ ≈ 73% of Barilla sales (SKU concentration)
- **Momentum panel:** Δ sales / share / penetration / frequency vs prior 13W

Q4 Promotion: Original Python finding (assignment)



Emerging/ Challenger Brands Radar:

💡 What the assignment couldn't show: who's coming next

Emerging = share < threshold, buyers ≥ floor, positive Δ share + (Δ penetration or Δ frequency), sustained 4 weeks.
Example shown at Max share = 10% to include challenger brands; at 5% the radar narrows to only the smallest emergers.

⭐ Barilla Plus — strongest signal

+1.61pp penetration, low promo (11%) → organic momentum

★ Ronzoni — quiet climb

Penetration-led gain (+1.20pp), promo ~0

⭐ Mueller — weak signal

Small Δ share +0.13pp, frequency not improving

Emerging Brands Radar

Definition: Emerging = share < threshold, buyers ≥ floor, positive Δ share and (Δ penetration or Δ frequency), sustained 4 weeks | As-of week 104

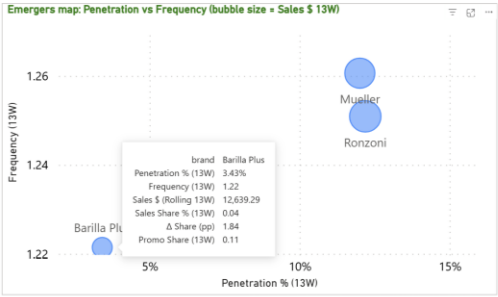
Share Threshold	Minimum Buyers	As-of Week
10.00%	1000	104

Candidates (this week)	Confirmed (Persist 4W)
4	3

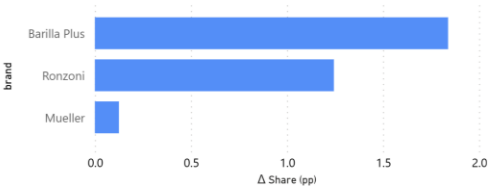
Confirmed Emerging Brands (Persist 4W)

brand	Sales Share % (13W)	Δ Share (pp)	Penetration % (13W)	Δ Penetration (pp)	Frequency (13W)	Δ Frequency	Promo Share (13W)	Score
Barilla Plus	0.04	1.84	3.43%	1.61	1.22	0.08	0.11	200.78
Ronzoni	0.10	1.24	12.19%	1.20	1.25	0.01	0.00	136.51
Mueller	0.09	0.13	12.01%	0.32	1.26	-0.01	0.00	15.69

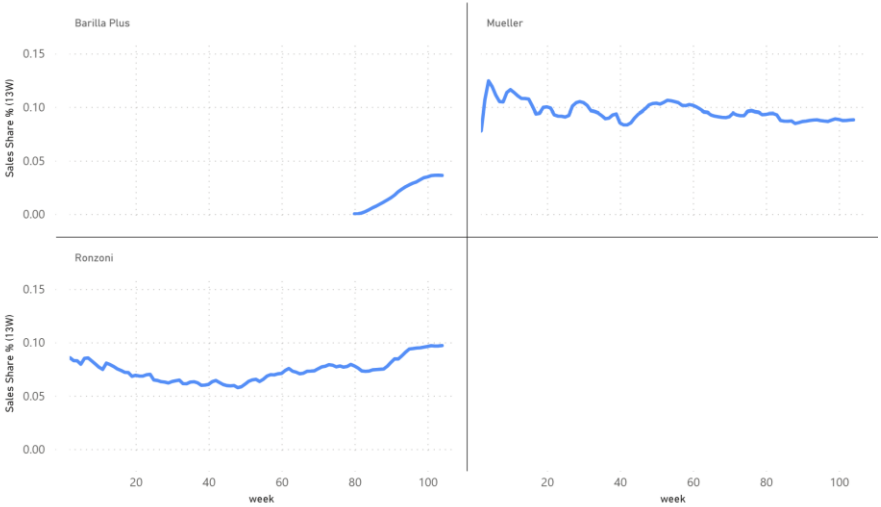
Emergers map: Penetration vs Frequency (bubble size = Sales $ 13W)



Δ Share (pp), last 13W vs prior 13W



Sales share trend (Rolling 13W) vs confirmed Emergers



From Static analysis → reusable monitoring tool.
Change threshold. change week.